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CERTIFICATE OF PANEL APPROVAL

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ABSTRACT

Type your abstract between the dashed lines. For a sentence example, did you know that the phrase “the quick brown fox jumps over the lazy dog” contains every letter of the English alphabet?

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ACKNOWLEDGMENT

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LIST OF NOTATIONS

Notation	Description	Page
$\mathbb{R}$	set of real numbers	4
$\mathbb{R}^+$	set of positive real numbers	4
$\mathbb{R}^-$	set of negative real numbers	4
$\mathbb{N}$	set of natural numbers	5
$a * b$	image of (a, b) under a binary operation $*$	7
e_G	identity element in a group G	7
a^{-1}	inverse element of an element a in a group	7
$V(G)$	vertex set of a graph G	9
$E(G)$	edge set of a graph G	9

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CHAPTER 1

INTRODUCTION

Introductory paragraph for Chapter 1.

1.1 Background of the Study

Insert background of the study here.

As always, an indented new paragraph is written by just inserting a blank line between paragraphs.

There are two types of in-text citations in APA referencing style. The first type is for referencing the paper in which the statement is taken. For example, consider the following sentences.

1. Every generalized topology is generated by a monotonic function (Császár, 2005).

2. If a function is differentiable at $x_0 \in \mathbb{R}$, then it is also continuous at x_0 (Bartle & Sherbert, 2011).

The second type is for referencing the paper in which the author(s) is/are part of the statement, as seen in the following sentences.

1. Further details of this result can be found in the dissertation of Ng (2000).

2. Barasara et al. (2011) introduced the concept of the shadow graph of a

graph.

The first type uses the command \parencite{key}, while the second type uses the command \textcite{key}, where key is the citation key of the reference material from the references.bib file. Note that there is no need to type the name of the author(s), because the compiler will automatically type the author's name according to the APA bibliography style.

If a specified result within the reference is cited, e.g., Theorem 4.1 of the dissertation of Ng, using the first type of citation, we refer to it like this: (Ng, 2000, Theorem 4.1). Here, the command \parencite[ref]{key} is used, where ref is the cited result. For the second type of citation, we use the command \textcite[ref]{key}, e.g., "see the result from the dissertation of Ng (2000, Theorem 4.1)".

1.2 Statement of the Problem

Insert statement of the problem here.

1.3 Objectives of the Study

Insert objectives here.

1.4 Significance of the Study

Insert significance here.

1.5 Scope and Limitation

Insert scope and limitation here.

1.6 Methodology

Insert methodology here.

CHAPTER 2

PRELIMINARIES

Introductory paragraph for Chapter 2.

2.1 Section 1

Introductory paragraph for Section 1.

Definition 2.1.1 (Bartle & Sherbert, 2011) Let $\mathbb{R}$ denote the set of real numbers. A *successor set* is a subset S of $\mathbb{R}$ with the following properties:

(i) $1 \in S$; and

(ii) If $n \in S$, then $n + 1 \in S$.

Example 2.1.2 The set $\mathbb{R}^+$ of positive real numbers is a successor set, since it satisfies properties (i) and (ii) of Definition 2.1.1. The set $\mathbb{R}^-$ of negative real numbers is not a successor set, since $1 \notin \mathbb{R}^-$. The open interval $(0, 10)$ is not a successor set, because $9 \in (0, 10)$ but $9 + 1 = 10 \notin (0, 10)$.

Proposition 2.1.3 *The intersection of any collection of successor sets is a successor set.*

Proof: Let $\mathcal{S}$ be a collection of successor sets, and set $T := \bigcap_{S \in \mathcal{S}} S$. We will show that T is a successor set by verifying properties (i) and (ii) of Definition 2.1.1 for the set T.

(i) Because each set $S \in \mathcal{S}$ is a successor set, $1 \in \mathcal{S}$ for all $S \in \mathcal{S}$. Thus, $1 \in T$.

(ii) Let $n \in T$. Then $n \in S$ for all $S \in \mathcal{S}$. Since each set $S \in \mathcal{S}$ is a successor set, it follows that $n + 1 \in S$ for all $S \in \mathcal{S}$. Hence, $n + 1 \in T$.

Therefore, T is a successor set. □

Definition 2.1.4 (Bartle & Sherbert, 2011) The set of *natural numbers*, denoted by $\mathbb{N}$, is defined to be the intersection of all successor sets of $\mathbb{R}$. In symbols,

$$\mathbb{N} := \bigcap \{S \subseteq \mathbb{R} : S \text{ is a successor set}\}.$$

Proposition 2.1.3 guarantees that $\mathbb{N}$ is a successor set. In fact, by definition, $\mathbb{N}$ is the "smallest" among all successor sets of $\mathbb{R}$ in the sense that $\mathbb{N} \subseteq S$ for all successor sets S.

From here, we prove the following result that is used to prove statements involving natural numbers. See the book of Bartle and Sherbert (2011).

Theorem 2.1.5 (Principle of Mathematical Induction) *Let P_n be a statement involving $n \in \mathbb{N}$. Suppose that the following conditions hold:*

(PM1) *P_1 is true.*

(PM2) *For $k \in \mathbb{N}$, if P_k is true, then P_{k+1} is true.*

Then P_n is true for all $n \in \mathbb{N}$.

Proof: Let $S := \{m \in \mathbb{N} : P_m \text{ is true}\}$. The conclusion of the theorem follows if we can prove that $S = \mathbb{N}$. By definition of S, we have $S \subseteq \mathbb{N}$, so it is enough to show that $\mathbb{N} \subseteq S$.

For this, we claim that S is a successor set. Since P_1 is true by **(PM1)**, $1 \in S$. Suppose that $n \in S$. Then P_n is true. By **(PM2)**, P_{n+1} is true, so that $n + 1 \in S$. Thus, S is a successor set.

Since $\mathbb{N}$ is the intersection of all successor sets, it follows that $\mathbb{N} \subseteq S$. Therefore, $S = \mathbb{N}$. □

Example 2.1.6 Prove that for all $n \in \mathbb{N}$,

$$1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}. \tag{2.1}$$

Proof: For $n \in \mathbb{N}$, let P_n be statement (2.1). Then P_1 is the statement

$$1^3 = \frac{1^2(1+1)^2}{4}, \tag{2.2}$$

which is true because both sides of Equation (2.2) are equal to 1. To finish the proof using Theorem 2.1.5, it suffices to prove **(PM2)** for the statement P_n. Suppose that for $k \in \mathbb{N}$, P_k is true, that is,

$$1^3 + 2^3 + \cdots + k^3 = \frac{k^2(k+1)^2}{4}. \tag{2.3}$$

We claim that P_{k+1} is true, that is,

$$1^3 + 2^3 + \cdots + (k+1)^3 = \frac{(k+1)^2(k+2)^2}{4}.$$

Indeed, by using (2.3) and algebraic manipulation, we have

$$\begin{aligned}
1^3 + 2^3 + \cdots + (k+1)^3 &= \big(1^3 + 2^3 + \cdots + k^3\big) + (k+1)^3 \\
&= \frac{k^2(k+1)^2}{4} + (k+1)^3 \\
&= \frac{k^2(k+1)^2 + 4(k+1)^3}{4} \\
&= \frac{(k+1)^2\big(k^2 + 4(k+1)\big)}{4} \\
&= \frac{(k+1)^2\big(k^2 + 4k + 4\big)}{4} \\
&= \frac{(k+1)^2(k+2)^2}{4}.
\end{aligned}$$

□

2.2 Section 2

Introductory paragraph for Section 2.

Definition 2.2.1 A *group* is a set G together with a map $*$ from $G \times G$ into G (where for $a, b \in G$, the image of (a, b) under $*$ is denoted by $a * b$), called a *binary operation* on G, such that the following hold.

(G1) The binary operation $*$ is *associative*:

$$(a * b) * c = a * (b * c), \quad \text{for all } a, b, c \in G.$$

(G2) There exists an element in G called the *identity* in G under $*$, denoted by e (or e_G if G needs to be indicated), such that $a * e = e * a = a$ for all $a \in G$.

(G3) For each $a \in G$, there exists an element in G called the *inverse* of a in G under $*$, denoted by a^{-1} , such that $a * a^{-1} = a^{-1} * a = e$.

Example 2.2.2 Let $G := \{u, v, w, x, y, z\}$ and define a binary operation $*$ on G as follows: for $a, b \in G$, $a * b$ is defined to be the element in G on Table 2.1 located in the same row as that of the element a before the vertical line and in the same column as that of the element b above the horizontal line. For instance, according to the table, we have $w * y = x$ and $y * w = v$. The fact that there is exactly one element in G in each entry on the lower right block of the table ensures that $*$ is a well-defined binary operation on G. This table is called a *Cayley table* for the binary operation $*$.

Table 2.1: A Cayley table for a binary operation $*$ on the set $\{u, v, w, x, y, z\}$

$*$	u	v	w	x	y	z
u	u	v	w	x	y	z
v	v	u	y	z	w	x
w	w	z	u	y	x	v
x	x	y	z	u	v	w
y	y	x	v	w	z	u
z	z	w	x	v	u	y

We will show that G is a group under $*$. Note that $*$ is associative. Indeed, we can verify using Table 2.1 that

$$(w * x) * z = y * z = u = w * w = w * (x * z),$$

and similar computations hold for all other choices of elements in G. Next, the identity element in G under $*$ is u, because $v * u = u * v = v$, and similar statements hold for the other elements in G. Finally, since $y * z = z * y = u$, we have $y^{-1} = z$ and $z^{-1} = y$, and a quick inspection of Table 2.1 shows that $u^{-1} = u$, $v^{-1} = v$, $w^{-1} = w$, and $x^{-1} = x$. Therefore, G is a group under $*$.

2.3 Section 3

Introductory paragraph for Section 3. All definitions are taken from Chartrand and Lesniak (2005).

Definition 2.3.1 A *graph* G is a pair $(V(G), E(G))$ consisting of a nonempty finite set $V(G)$ and a set $E(G)$ of unordered pairs of elements of $V(G)$. The sets $V(G)$ and $E(G)$ are called the *vertex set* and the *edge set* of the graph G, respectively. The elements of $V(G)$ are called *vertices* in G, while the elements of $E(G)$ are called *edges* in G. The cardinalities of $V(G)$ and $E(G)$ are called the *order* and the *size* of G, respectively.

Usually, a graph is represented visually by drawing a point or circle for every vertex of the graph, and drawing a curve (usually a line segment) connecting two vertices, say, v and w, if the unordered pair $\{v, w\}$ (written briefly as vw) is an edge of the graph.

Example 2.3.2 Figure 2.1 is an illustration of a graph G with vertex set $V(G) := \{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$ and edge set $E(G) := \{v_1v_3, v_1v_4, v_2v_4, v_3v_4, v_4v_5, v_4v_6, v_4v_7, v_5v_6\}$. In this case, the order of G is 7, while its size is 8.

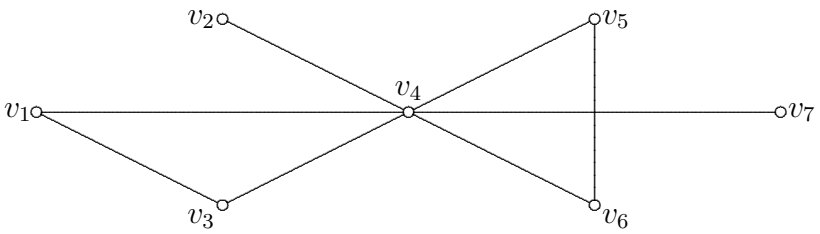


Figure 2.1: A graph with 7 vertices and 8 edges

CHAPTER 3

MAIN RESULTS

Introductory paragraph for Chapter 3.

3.1 Main Results 1

Main result 1 discussion here...

3.2 Main Results 2

Main result 2 discussion here...

CHAPTER 4

SUMMARY AND RECOMMENDATIONS

4.1 Summary

Insert summary here.

4.2 Recommendations

Insert recommendations here.

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APPENDIX A

TITLE OF APPENDIX A

Introduction to Appendix A here.

A.1 Appendix A Section 1

Appendix A Section 1 discussion here.

APPENDIX B

TITLE OF APPENDIX B

Introduction to Appendix B here.

B.1 Appendix B Section 1

Appendix B Section 1 discussion here.

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I also declare that the intellectual content of this manuscript is the product of my own work, except to the extent that assistance from others in the project's design and conception or in style, presentation and linguistic expression is acknowledged.

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